

BUSINESS ANALYTICS PROJECT PORTFOLIO REPORT

Project Title:

Rental Market Intelligence & Investment Decision Framework

Market Focus:

Abu Dhabi Real Estate Rental Market Analysis

Technology Stack:

Microsoft Excel | SQL Server | Power BI | DAX

SECTION 1: BUSINESS REQUIREMENT DOCUMENT (BRD)

1.1 Problem Statement

Right now, real estate investors, brokers, and property consultants in Abu Dhabi have a hard time checking if a rental property is actually a good deal. Most agencies rely on gut feeling, scattered spreadsheets, or incomplete property lists because they do not have a shared, data-backed system.

This manual approach causes a few major problems in everyday business operations:

- **No Clear Standards:** It is hard to compare different neighbourhoods side-by-side to see where rent is actually fair compared to the physical size of the property.
- **Overpricing and Missed Opportunities:** Agents often market properties that are way overpriced without realizing it, or they miss out on great, underpriced deals that would give their clients an excellent return on investment (ROI).
- **Stagnant Listings:** Properties sit on the market for months without anyone tracking them. Because teams do not look at how old an ad is, companies waste their advertising budgets running ads for apartments that are either already taken or completely uncompetitive.

1.2 Strategic Business Objectives

The main goal of this project is to stop guessing and start using data to drive property decisions. By building this system, we want to help the platform achieve these specific goals:

Make Better Recommendations: Improve investment decisions by providing reliable rental market benchmarks, pricing insights, and performance indicators.

Find the Best Areas: Automatically rank neighbourhoods based on true price-to-size efficiency (Rent per Square Foot) rather than just looking at the total rent.

Save Time: Speed up the consultation process. Instead of taking days to manually research and compare properties for an investor, an advisory agent can cross-filter the parameters and deliver a reliable market benchmark in seconds.

1.3 Target Operational Stakeholders

Property Investors (HNI / NRI Clients): They need low-risk areas with proven high returns to invest their money safely.

Real Estate Consultants: They need instant proof of average market prices so they can negotiate better lease terms for their corporate clients.

Brokerage Sales Teams: They need real data and clear stories to convince leads to buy properties.

Property Management Firms: They need to find slow-moving listings early so they can fix marketing plans before properties become dead stock.

1.4 Core Key Performance Indicators (KPIs)

Rent per Sqft (Pricing Efficiency): This tracks exactly how much rent a property generates for each square foot of space. It is calculated by dividing total rent by the area size.

Listing Age (Market Velocity): This measures how many days a property ad has been live on the market since it was first posted.

Demand Index: This measures how fast a listing moves. A low average listing age means a high demand index, showing that properties in that area are highly desirable and selling fast.

SECTION 2: FUNCTIONAL REQUIREMENT DOCUMENT (FRD)

2.1 Functional Overview

The system provides an interactive Business Intelligence dashboard that enables users to analyse Abu Dhabi rental property data.

The solution combines data preparation, SQL analysis, and Power BI visualization to deliver actionable insights.

2.2 Data Requirements

The dashboard uses rental property data containing:

- Property Address
 - City
 - Location
 - Property Type
 - Bedrooms
 - Bathrooms
 - Area (Sqft)
 - Annual Rent
 - Rent per Sqft
 - Furnishing Status
 - Listing Age
 - Latitude
 - Longitude
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2.3 Dashboard Requirements

2.3.1 Executive Dashboard

Features:

- Total Portfolio Value
- Average Annual Rent
- Total Active Listings
- Average Rent per Sqft
- Demand Indicators
- KPI Monitoring

2.3.2 Location Intelligence Dashboard

Features:

- Geographic rental analysis
- Location performance comparison
- High-demand areas identification
- Rental distribution by community

2.3.3 Rental Analysis Dashboard

Features:

- Rental price comparison
 - Property type analysis
 - Bedroom and area analysis
 - Furnishing impact analysis
 - Listing age analysis
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2.4 Data Processing Requirements

The solution should support:

- Data cleaning
 - Data transformation
 - Data validation
 - Data quality checks
 - Data modeling
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2.5 Technical Requirements

Tools Used:

- Microsoft Excel
 - SQL Server
 - Power BI
 - DAX
 - GitHub
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2.6 User Requirements

Users should be able to:

- Filter data interactively
 - Compare properties
 - Analyze rental trends
 - Identify investment opportunities
 - Export insights for decision-making
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2.7 Performance Requirements

The dashboard should:

- Load efficiently
 - Provide accurate calculations
 - Display interactive visuals
 - Support multiple analytical views
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2.8 Success Criteria

The project is successful when users can:

- Understand rental market trends
 - Identify valuable locations
 - Evaluate property performance
 - Make data-driven investment decisions
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SECTION 3: BUSINESS QUESTIONS & ANALYSIS ANSWERS

3.1 Which locations have the highest rental demand?

Answer:

The analysis identifies locations with higher rental activity by evaluating:

- Number of active listings
- Average annual rent
- Rental price efficiency

- Listing concentration

High-demand locations show a strong combination of property availability, rental value, and market activity. The location intelligence dashboard helps stakeholders compare rental performance across different communities.

3.2 What factors influence rental pricing?

Answer:

Rental pricing is influenced by multiple property attributes:

- Location
- Property type
- Area size (Sqft)
- Number of bedrooms
- Furnishing status
- Market demand

Properties in premium locations and larger property categories generally show higher rental values.

3.3 Which property types provide better investment opportunities?

Answer:

Property investment opportunities are evaluated based on:

- Rental income potential
- Rent per Sqft
- Demand indicators
- Listing performance

The analysis helps identify property categories that provide better rental efficiency and investment potential.

3.4 How does rental price vary across Abu Dhabi communities?

Answer:

Rental prices vary significantly across locations due to:

- Community popularity
- Accessibility
- Property characteristics
- Demand levels

The dashboard provides location-based comparisons using geographic visualization and rental KPIs.

3.5 Which properties are underpriced or overpriced?

Answer:

Properties are evaluated by comparing:

- Annual rent
- Area size
- Rent per Sqft
- Location averages

Properties with lower rent per Sqft compared to similar properties may represent potential investment opportunities, while significantly higher values may indicate premium pricing.

3.6 What is the impact of furnishing status on rental value?

Answer:

Furnished properties generally demonstrate different rental pricing patterns compared with unfurnished properties.

The analysis compares rental values by furnishing category to understand its impact on rental performance.

3.7 How can investors use this dashboard?

Answer:

Investors can use the dashboard to:

- Identify profitable locations
- Compare rental performance
- Evaluate property opportunities

- Understand market trends
- Make data-driven investment decisions

SECTION 4: KEY BUSINESS INSIGHTS

- 4.1 High-Performing Rental Locations
- 4.2 Rental Efficiency Analysis
- 4.3 Property Characteristics Impact
- 4.4 Market Demand & Listing Performance
- 4.5 Power BI Dashboard Contribution

SECTION 5: DATA QUALITY AUDIT & OPERATIONAL ANOMALIES LOG

During data quality assessment, three technical issues were identified that required structural improvements to ensure reliable database calculations and dashboard performance.

5.1 Data Type Optimization & Numeric Overflow Issue

While working with the raw dataset, a numeric aggregation issue was identified. The total rental value across 73,742 records exceeded standard integer limits, which could create overflow risks during SQL calculations.

Solution:

Rental calculations were optimized by applying appropriate numeric data types and using CAST(Rent AS FLOAT) during aggregation to ensure accurate calculations.

5.2 Listing Demand & Marketing Efficiency Issue

The analysis identified properties with extended listing periods and low demand indicators. Long-standing listings may represent inefficient marketing spend and require further evaluation.

Solution: Listing age and demand indicators were incorporated into the dashboard to help stakeholders identify slow-moving properties and improve marketing decisions.

5.3 Spatial Data Quality & Missing Area Values Issue

The dataset contained records where rental prices were available but property area values were recorded as zero. These records could create calculation errors during Rent per Sqft analysis.

Solution: Data validation checks were applied to identify and handle invalid area values before dashboard calculations and visualizations were performed.

SECTION 6: STRATEGIC ADVISORY & RECOMMENDATIONS

6.1 Enforce Dynamic Pricing Strategy: Real estate consultants should use the gap analysis chart during landlord negotiations to enforce pricing boundaries, advising asset positioning at or below the market average line to capture rapid conversion velocity.

6.2 Reduce Marketing Capital Leakage: Implement an automated business rule that immediately flags and re-evaluates active property ad spend once a listing crosses the 60-day market threshold without transaction movement, eliminating wasted pay-per-click budget on stagnant stock.

6.3 Improve Client Advisory Efficiency: Leverage the unified 3-page reporting layout to replace manual research. Agents can utilize the grouped sidebar control panel to answer high-intent investor valuation and location queries instantly, reducing manual research effort and improving the speed of investment analysis.

SECTION 7: FINAL PROJECT SUMMARY & CONCLUSION

7.1 Project Outcome

This project successfully developed an end-to-end **Rental Market Intelligence and Investment Decision Framework** by transforming raw real estate data into meaningful business insights using **Microsoft Excel, SQL Server, and Power BI**.

The analysis covered **73,742+ Abu Dhabi rental property records** and focused on understanding rental market trends, property characteristics, location performance, pricing behavior, and investment opportunities. The project involved data quality assessment, data cleaning, SQL-based analysis, KPI development, and interactive dashboard creation.

The final Power BI solution provided a structured analytical platform consisting of:

- Executive-level KPI monitoring
- Rental market performance analysis
- Geographic location intelligence
- Rental price and property trend analysis
- Investment opportunity evaluation

The dashboard enables stakeholders to explore rental patterns, compare locations, identify high-performing areas, and support data-driven real estate decisions.

7.2 Business Value Delivered

The project delivered significant business value by converting complex property data into actionable insights for investors, real estate professionals, and decision-makers.

Key business benefits include:

Improved Investment Decision-Making

- Identified high-demand locations based on rental performance, property availability, and pricing trends.
- Supported investors in evaluating potential rental investment opportunities using measurable indicators.

Enhanced Market Understanding

- Provided visibility into rental price variations across different locations and property categories.
- Highlighted relationships between property size, amenities, location, and rental value.

Data-Driven Strategic Planning

- Developed KPIs such as average rent, rent per square foot, property distribution, and demand indicators to measure market performance.
- Enabled faster comparison of locations and property segments.

Operational Efficiency

- Reduced dependency on manual analysis by creating an interactive dashboard solution.
- Established a reusable reporting framework that can support future rental market monitoring.

Data Quality Improvement

- Identified and documented data issues including missing values, inconsistent formats, and potential calculation risks.
 - Applied validation and cleaning approaches to improve reliability of analytical outputs.
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7.3 Future Enhancement Opportunities

Although the current solution provides valuable market insights, several enhancements can further improve its analytical capability and business impact.

1. Predictive Rental Price Forecasting

Implement machine learning models using historical rental trends, location factors, and property characteristics to forecast future rental prices.

2. Real-Time Data Integration

Connect the dashboard with live property listing sources or APIs to provide continuously updated market insights.

3. Advanced Investment Scoring Model

Develop an investment ranking framework combining:

- Rental yield
- Demand index
- Price appreciation potential
- Location attractiveness
- Property characteristics

to automatically identify the best investment opportunities.

4. Geospatial Analytics Enhancement

Integrate advanced mapping features to analyse:

- Distance from business hubs
- Transportation accessibility
- Nearby facilities
- Area growth patterns

5. Automated Reporting

Implement Power BI subscriptions, Power Automate workflows, or scheduled reporting systems to deliver automated market updates to stakeholders.

6. Expansion to Sales Market Analysis

Extend the framework beyond rentals by incorporating property sales data to analyse:

- Capital appreciation
- Market cycles
- Buyer demand
- Long-term investment potential

Conclusion

This project demonstrates how **Business Analytics and Business Intelligence tools can transform large-scale real estate datasets into strategic decision-making solutions.**

By combining SQL data processing, Excel-based validation, and Power BI visualization, the project created a comprehensive analytical framework that improves market transparency, supports investment evaluation, and enables evidence-based decision-making.

The developed solution provides a strong foundation for future enhancements such as predictive analytics, automation, and advanced real estate intelligence capabilities.